LungLogic Single-Lung Flow Control

PID, Actuator Mapping, Smith Predictor, and Calibration Experiments

Group Meeting

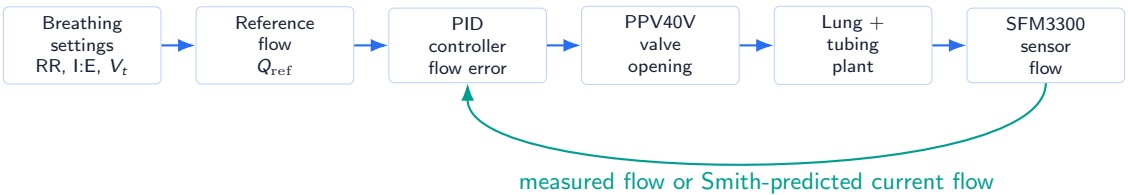
Presentation Focus

- Where is PID used in the current code?
- What physical variable does PID control?
- What is the actuator?
- What does the Smith predictor do in the loop?
- What parameters does PID need, and how should they be calibrated?
- What parameters does the Smith predictor need, and how should they be calibrated?

Key Message

PID controls flow rate. The continuous actuator is the PPV40V proportional pinch valve. The Smith predictor only changes the feedback signal seen by PID to compensate pneumatic and sensor delay.

Closed-Loop Structure



- Loop objective: make measured flow Q_{measured} track reference flow Q_{ref}.
- Volume is not directly controlled by PID. It is obtained by integrating flow and is used as a phase stop condition.

Where PID Is Used

- Python simulation: `single_lung_pid_gui.py`, `PIDController.update()`.
- Simulation loop: `run_single_lung_simulation()` calls PID during inspiration and expiration.
- Hardware path: `single_lung_hardware_gui.py` sends K_p, K_i, K_d and related parameters to Arduino through `CFG`.
- Arduino firmware: `updatePID()` runs inside the 100 Hz control loop.
- PID is reset at state transitions to avoid carrying integral error across breathing phases.

PID Control Law

$$a = \mathrm{clip}\left(a_{\mathrm{ff}} + K_p e + K_i \int e\,dt + K_d d,\ 0,\ 1\right)$$

What PID Controls

Controlled Variable

$$Q_{\text{measured}} \rightarrow Q_{\text{ref}}$$

PID controls **flow rate**, not pressure and not volume directly.

Inspiration Reference

$$Q_{\text{in,ref}} = 0.06 \frac{V_{\text{target}}}{T_{\text{fill}}}$$

Default 250 mL / 1.30 s:

$$Q_{\text{in,ref}} = 11.54 \text{ SLM}$$

Expiration Reference

$$Q_{\text{out,ref}} = -0.06 \frac{V_{\text{target}}}{T_{\text{usable,out}}}$$

PID tracks the magnitude of negative expiration flow.

Volume Estimate

$$V[k] = V[k-1] + Q[k] \frac{1000}{60} \Delta t$$

Volume is used to determine whether the target tidal volume has been reached.

Actuators

Continuous Control Actuator

- PPV40V proportional pinch valve
- PID outputs normalized opening $a \in [0, 1]$
- $a = 0$ means closed, $a = 1$ means fully open

Voltage Mapping

$$U_{\text{cmd}} = 5(1 - a)$$

$$a = 1 \Rightarrow 0 \text{ V fully open}$$

$$a = 0 \Rightarrow 5 \text{ V fully closed}$$

Direction Selection Actuators

- Atmosphere solenoid valve
- Vacuum solenoid valve

State Machine Output

- Inspiration: atmosphere valve open, vacuum valve closed
- Expiration: atmosphere valve closed, vacuum valve open
- Dead time: both directional valves closed

PID continuously modulates PPV40V; solenoid valves only switch the atmosphere/vacuum path.

Why Smith Predictor Is Needed

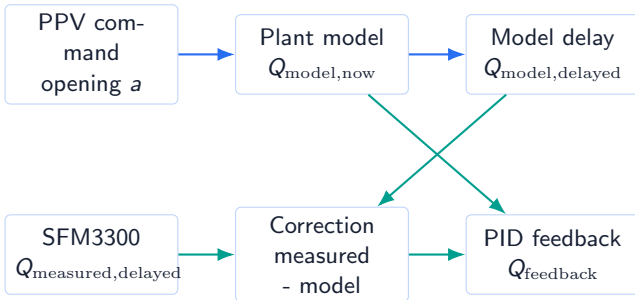
- Pneumatic tubing introduces transport delay.
- SFM3300 sensing and signal processing introduce response lag.
- If PID directly uses delayed feedback, the response may overshoot, oscillate, or require conservative gains.
- The Smith predictor uses the model to estimate current flow, then corrects model error using delayed real measurement.

Smith Predictor Feedback Estimate

$$Q_{\text{feedback}} = Q_{\text{model,now}} + g\left(Q_{\text{measured,delayed}} - Q_{\text{model,delayed}}\right)$$

The Smith predictor does not replace PID; it provides delay-compensated flow feedback to PID.

Smith Predictor Position



- Smith disabled: PID feedback = measured flow.
- Smith enabled: PID feedback = predicted current flow.
- Controller and actuator are unchanged; only the feedback signal changes.

Current Model Parameters

Lung and Valve Model

Parameter	Current value	Meaning
`capacity_ml`	250 mL	effective lung capacity
`tau_in_s`	0.57 s	inspiration time constant
`tau_out_s`	0.50 s	expiration time constant
`valve_exponent`	1.15	PPV nonlinearity exponent
`transport_delay_s`	0.025 s	pneumatic transport delay
`sensor_tau_s`	0.020 s	sensor response time constant
`sensor_noise_slm`	0.10 SLM	simulated sensor noise
`flow_sign`	1	sensor direction convention

These are initial values for simulation and early hardware tuning, not final identified parameters.

Current Controller Parameters

PID Parameters

Parameter	Current value
K_p	0.022
K_i	0.16–0.21
K_d	0.0
integral limit	2.0
control period	0.01 s

Smith Parameters

Parameter	Current value
enabled	false by default
delay	0.025 s
correction gain	0.85–1.0

Code Status

PID is active in the flow loop. Smith predictor is implemented and configurable, but disabled by default in the current GUI config.

Parameters Needed by PID

PID Controller Parameters

- K_p: proportional gain
- K_i: integral gain
- K_d: derivative gain
- integral limit: anti-windup bound
- Δt: control period

Feedforward Model Parameters

- $V_{\max}$: effective capacity
- τ_{in}: inspiration time constant
- τ_{out}: expiration time constant
- γ: valve nonlinearity exponent

Feedforward Opening

$$a_{\text{ff}} = \left(\frac{Q_{\text{ref}}}{Q_{\max}} \right)^{1/\gamma}$$

$$Q_{\max,\text{in}} = 0.06 \frac{V_{\max} - V}{\tau_{\text{in}}}, \quad Q_{\max,\text{out}} = 0.06 \frac{V}{\tau_{\text{out}}}$$

PID Calibration Experiments

1. **Open-loop PPV step tests**
   - Fix PPV opening at 20%, 40%, 60%, 80%, and 100%.
   - Test inspiration and expiration separately.
   - Record flow, integrated volume, and response time.
2. **Closed-loop PI tuning**
   - Start with $K_d = 0$.
   - Increase K_p until the response is fast but not strongly oscillatory.
   - Increase K_i to reduce steady-state flow error and tidal volume error.
3. **Multi-condition validation**
   - Test different RR, I:E, target volume, and target fill time.
   - Compare rise time, overshoot, steady-state error, and tidal volume error.

Parameters Needed by Smith Predictor

Smith-Specific Parameters

- L or `delay_s`: total delay
- g or `correction_gain`: correction gain
- `enabled`: whether Smith predictor is active

Most Important Parameter

$$L = t_{\text{flow response}} - t_{\text{command}}$$

This is the delay between command change and measured flow response.

Runtime Signals

- Current PPV opening a
- Pneumatic direction
- Delayed sensor measurement $Q_{\text{measured,delayed}}$
- Current model prediction $Q_{\text{model,now}}$
- Delayed model prediction $Q_{\text{model,delayed}}$

Correction Gain Meaning

$g = 0$: fully trust the model.
$g = 1$: fully apply delayed measurement error correction.
Too large g may amplify noise or model error.

Note

The Smith predictor reuses the first-order lung model already introduced in the PID/feedforward slide.

Smith Predictor Calibration Experiments

1. **Delay identification**
   - Apply a PPV opening step.
   - Record command timestamp, valve voltage, and SFM3300 flow.
   - Measure the time from command change to clear flow response to identify `delay_s`.
2. **Model-delay alignment**
   - Generate $Q_{\mathrm{model,now}}$ using the fitted model.
   - Delay the model output by L to obtain $Q_{\mathrm{model,delayed}}$.
   - Compare it against measured Q_{measured}.
3. **Smith on/off A-B test**
   - Keep control settings, breathing settings, and hardware state unchanged.
   - Run with Smith predictor disabled and enabled.
   - Compare overshoot, oscillation, time to target volume, and tidal volume error.

Basic Calibration Before Hardware Migration

- **SFM3300 zero-flow calibration**: close all valves and estimate flow offset and noise.
- **Flow direction calibration**: open atmosphere and vacuum paths separately, verify signs, and set `flow_sign`.
- **PPV40V voltage-flow mapping**: test multiple voltages and confirm 0 V fully open, 5 V fully closed.
- **Inspiration step response**: fit τ_{in} and γ.
- **Expiration step response**: fit τ_{out} and verify regulated vacuum range.
- **Safety validation**: emergency stop, flow limit, serial watchdog, and directional valve interlock.

Summary

- PID is used in the flow feedback loop.
- PID controls SFM3300 measured flow so it tracks Q_{ref}.
- PID outputs PPV40V proportional pinch valve opening.
- Solenoid valves only select the atmosphere or vacuum path.
- The Smith predictor estimates current flow and corrects model error using delayed measurement.
- PID calibration focuses on K_p, K_i, K_d, integral limit, and feedforward model parameters.
- Smith calibration focuses on total delay, correction gain, and enabled/disabled comparison.

One-Sentence Takeaway

State machine chooses pneumatic direction; PID modulates PPV opening; Smith predictor improves feedback timing.